

Clean Code: Error Handling (Complete Notes + Many Examples)

1. Use Exceptions Instead of Return Codes

Return codes clutter logic and are easy to ignore.

Bad Example:

```
if (handle == INVALID) {  
    return -1;  
}
```

Good Example:

```
if (handle == INVALID) {  
    throw new DeviceException("Invalid handle");  
}
```

2. Avoid Cluttered Error Handling

Error handling should not hide business logic.

Bad Example:

```
if (user != null) {  
    if (user.getProfile() != null) {  
        process(user);  
    }  
}
```

Good Example:

```
process(user);
```

3. Separate Logic and Error Handling

Keep try-catch at high level.

Bad Example:

```
try {  
    step1();  
    step2();  
} catch (Exception e) {  
    // everywhere  
}
```

Good Example:

```
try {  
    process();  
} catch (Exception e) {  
    handle(e);  
}
```

4. Avoid Checked Exception Cascade

Checked exceptions break encapsulation.

Bad Example:

```
public void a() throws Exception  
public void b() throws Exception
```

Good Example:

```
public void a()  
public void b()
```

5. Provide Context in Exceptions

Always explain what failed.

Bad Example:

```
throw new Exception("error");
```

Good Example:

```
throw new StorageException("Failed loading file: " + fileName);
```

6. Wrap Third-party API

Unify exceptions.

Bad Example:

```
try {  
    port.open();  
} catch(A e) {}  
catch(B e) {}
```

Good Example:

```
try {  
    port.open();  
} catch(DeviceFailure e) {}
```

7. Special Case Pattern

Avoid exception for normal flow.

Bad Example:

```
try {  
    total += get();  
} catch(NotFound e) {  
    total += 0;  
}
```

Good Example:

```
total += get();
```

8. Don't Return Null

Return empty object instead.

Bad Example:

```
if (list != null) {  
    for(Object o : list){}  
}
```

Good Example:

```
for(Object o : list){}
```

9. Don't Pass Null

Avoid null arguments.

Bad Example:

```
sendEmail(null);
```

Good Example:

```
sendEmail(user.getEmail());
```

10. Never Ignore Exceptions

Always log or rethrow.

Bad Example:

```
try {  
    read();  
} catch(Exception e) {}
```

Good Example:

```
catch(Exception e) {  
    logger.error("Read failed", e);  
}
```

11. Rethrow with Context

Don't swallow errors.

Bad Example:

```
catch(Exception e) {}
```

Good Example:

```
catch(Exception e) {  
    throw new ServiceException("Process failed", e);  
}
```